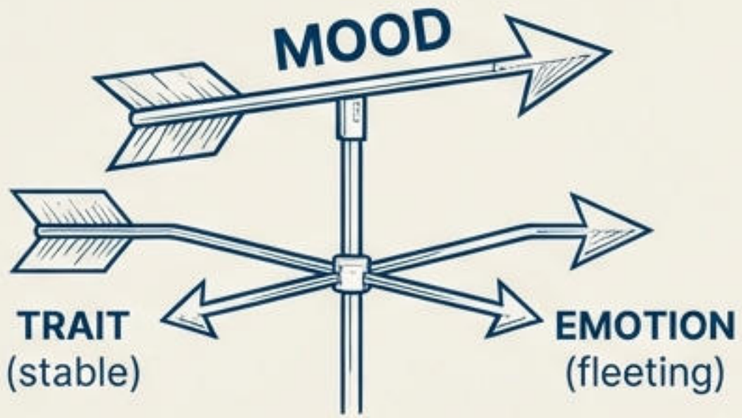
Mood vs. Emotion: A Comparative Analysis

In psychological science, the terms 'mood' and 'emotion' are often used interchangeably in everyday language, yet they represent distinct affective phenomena. Understanding their differences is fundamental to the study of affect.

A **mood** is best conceptualized as a diffuse, long-lasting, and subjectively experienced affective state. Unlike fleeting emotions or stable personality traits, moods form a kind of emotional background to our conscious experience, coloring our perception and cognition over extended periods—often for hours or even days.

This lecture will deconstruct the components of mood, exploring how these pervasive states are distinct from acute emotions, emotions, how they are scientifically measured, and the myriad factors—from our internal clocks to the changing seasons—that govern their subtle yet powerful fluctuations.



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Duration and Intensity

The most fundamental distinction between moods and emotions lies in their temporal and magnitudinal profiles.

Emotions are brief and intense affective episodes. They typically last for a matter of seconds to minutes at most. An emotional experience is characterized by a salient peak—a sharp, noticeable response to a specific trigger. When we speak of an emotion lasting for hours, we are more likely describing a series of recurrent emotional episodes, not a single, constant state.

Moods in contrast, are characterized by their persistence and lower intensity. They are more global, diffuse states that can last for hours or days. A mood doesn't have the sharp peaks of an emotion; rather, it's a sustained, low-level affective coloring. For example, ‘**irritability**’ is a mood, whereas a sudden burst of ‘**anger**’ is an emotion. If a state endures for weeks or months, it is classified as an **affective disorder**, not a mood.

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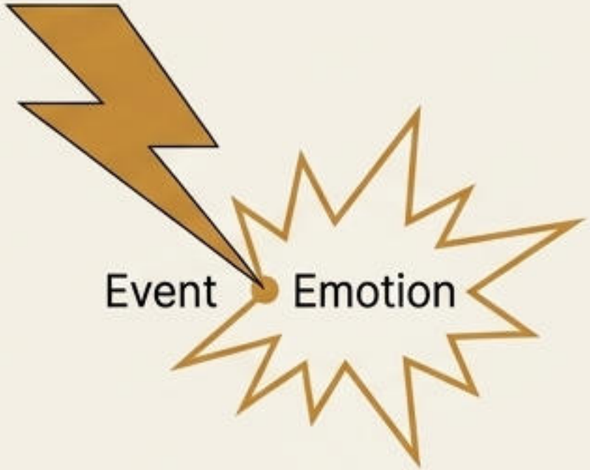
Awareness of Stimuli

The clarity and immediacy of the eliciting stimulus provide another key point of differentiation.

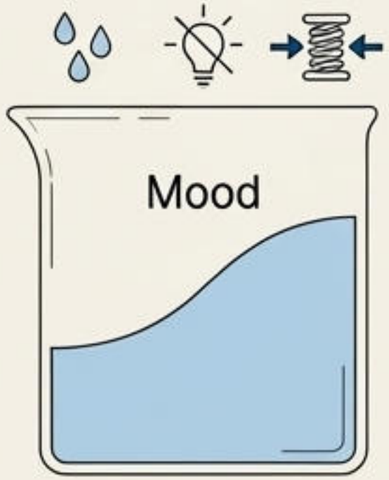
Emotions are typically provoked by a specific, identifiable object or event. The onset is often **sudden** and directly linked to a trigger, whether it is external (the punchline of a joke) or internal (a sudden memory). While we may not be consciously aware of the trigger *as it happens*, **we can usually specify the cause of an emotion in retrospect.**

Moods are more opaque in this regard. Their onset is often **gradual**, and individuals are frequently unaware of a specific cause. They seem to emerge without a clear, precipitating event. Moods can, however, be influenced by stimuli that unfold slowly over time, such as hormonal cycles, accumulating stress, or the gradual change of seasons.

Emotion Trigger



Mood Onset



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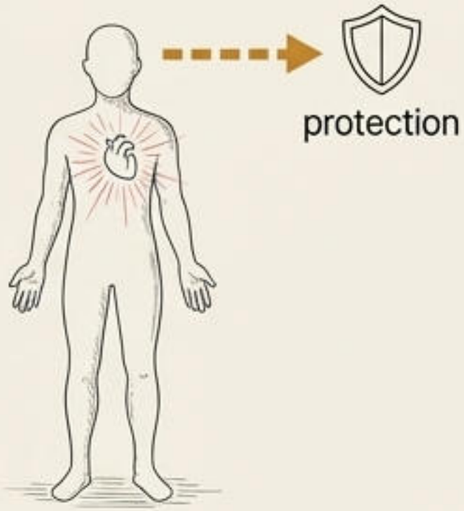
Consequences and Physiology

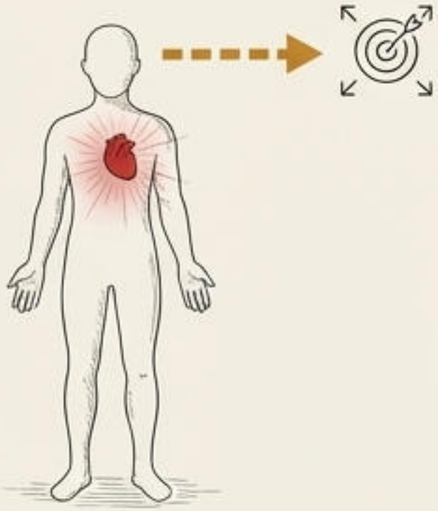
The functional role and physiological signature of moods and emotions also differ significantly.

Directionality. Emotions are often described as being directed toward an intended consequence; they have an aim. For example, the aim of fear is to promote avoidance of a threat, while jealousy may aim to protect a valued social relationship. Emotions prepare the body for a specific course of action. Moods, conversely, often feel directionless and without a specific intended outcome.

Physiological Salience. While both states involve physiological changes, the body's involvement is far more salient and consciously perceived during an emotional experience. The racing heart of fear or the heat of anger are potent physical sensations. The physiological underpinnings of a mood are typically more subtle and less accessible to conscious awareness.



Emotion



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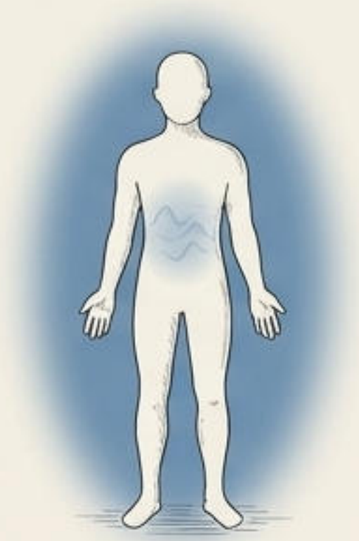

Mood

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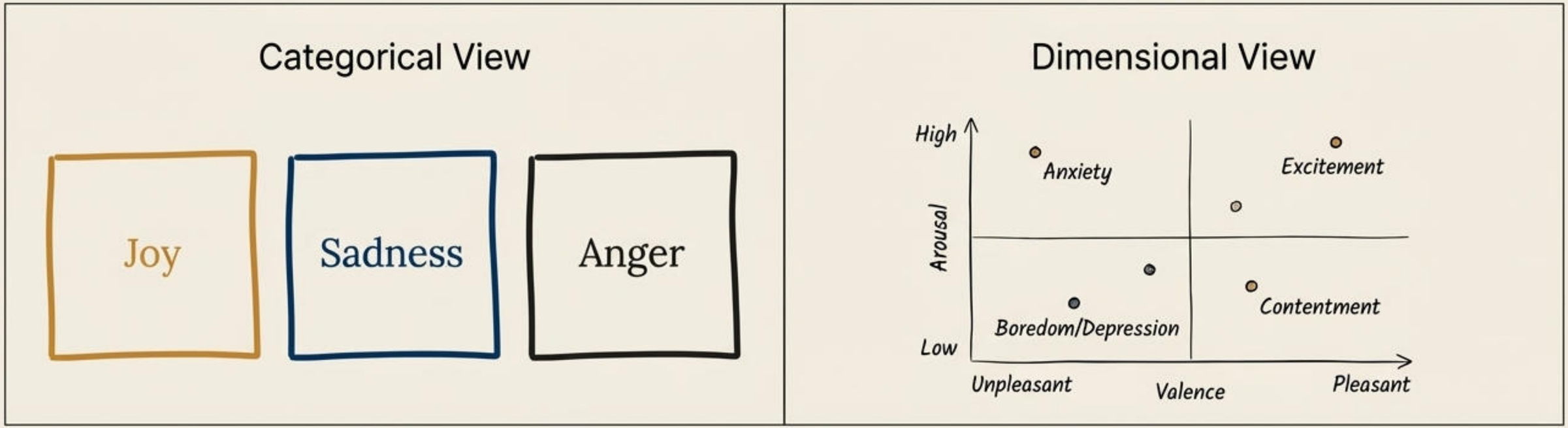
Measuring Affective Valence

Dimensional vs. Categorical Views

To study affect scientifically, we must first establish a conceptual framework for its structure. Two primary approaches dominate the field:

- **Categorical Approach:** This view, often applied to emotions, posits the existence of discrete, exclusive affective states. Rooted in the basic emotions tradition (e.g., happiness, sadness, anger, fear), this approach treats each emotion as a distinct category with a unique profile of appraisal, physiology, and expression.
- **Dimensional Approach:** This view proposes that affective states can be best understood by their position on a small number of underlying dimensions. For moods, the most critical dimensions are **valence** (pleasantness vs. unpleasantness) and **arousal** (activation vs. deactivation). In this framework, moods are not distinct categories but rather have a positive or negative feeling that can vary in intensity. This dimensional view is foundational to most modern mood measurement.



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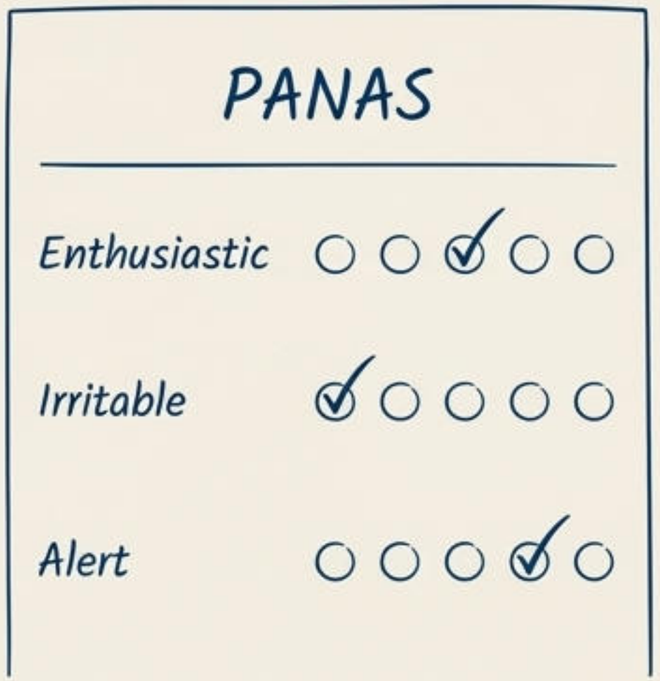
The Positive and Negative Affect Schedule (PANAS)

Dimensional vs. Categorical Views

The dimensional approach led to the development of one of the most widely used instruments for mood assessment: the Positive and Negative Affect Schedule (PANAS). Developed by Watson, Clark, and Tellegen (1988), the PANAS was designed to provide brief, reliable, and valid measures of the two primary dimensions of mood.

The core insight of the PANAS is that positive and negative affect are not bipolar opposites on a single continuum, but rather are two distinct, largely independent dimensions. A person can be high in both, low in both, or high in one and low in the other.

The scale asks participants to rate the extent to which they have experienced a series of affective states over a specified time frame (e.g., "right now," "the past week," "in general"). This flexibility allows it to measure both transient mood states and more stable affective traits.



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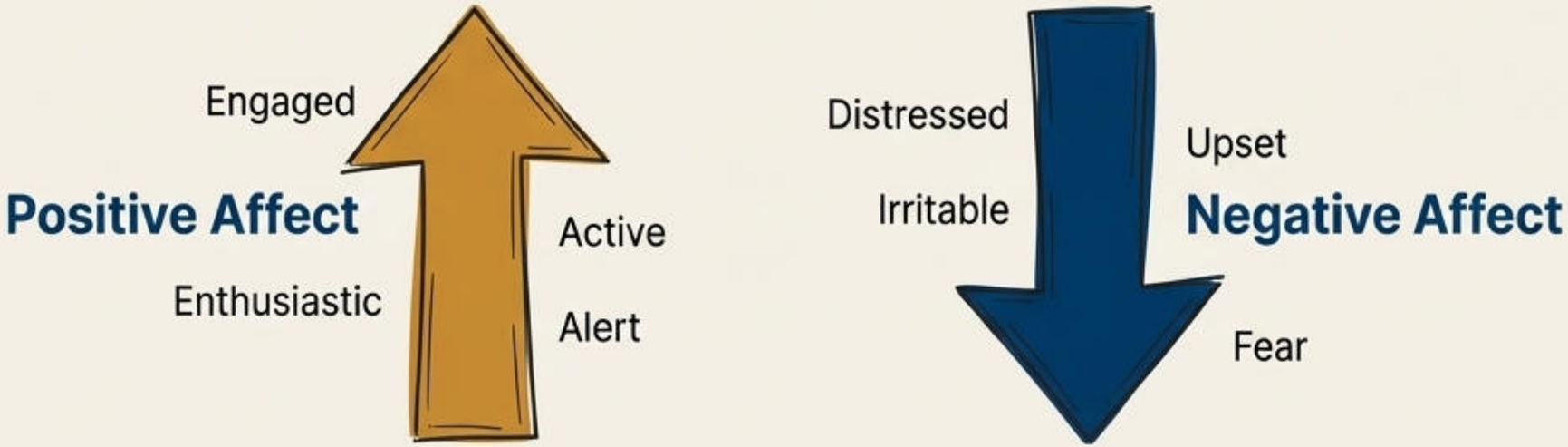
Defining the Two Dimensions

The PANAS measures two broad, higher-order factors of emotional experience:

Positive Affect (PA):
This dimension reflects the extent to which an individual feels **enthusiastic, active, determined**, and **alert**. It is not simply the absence of negative feelings, but a state of pleasurable engagement, high energy, and full concentration. High PA is characterized by a tendency to approach and engage with situations and activities.

Negative Affect (NA): This dimension reflects **subjective distress** and unpleasurable engagement. It encompasses a variety of aversive mood states, including fear, irritability, guilt, and upset. High NA is characterized by feelings of distress and a tendency to withdraw or avoid situations.

The quasi-independence of these two factors is a critical feature, allowing for a more nuanced understanding of complex affective states like bittersweet nostalgia (co-occurrence of PA and NA).



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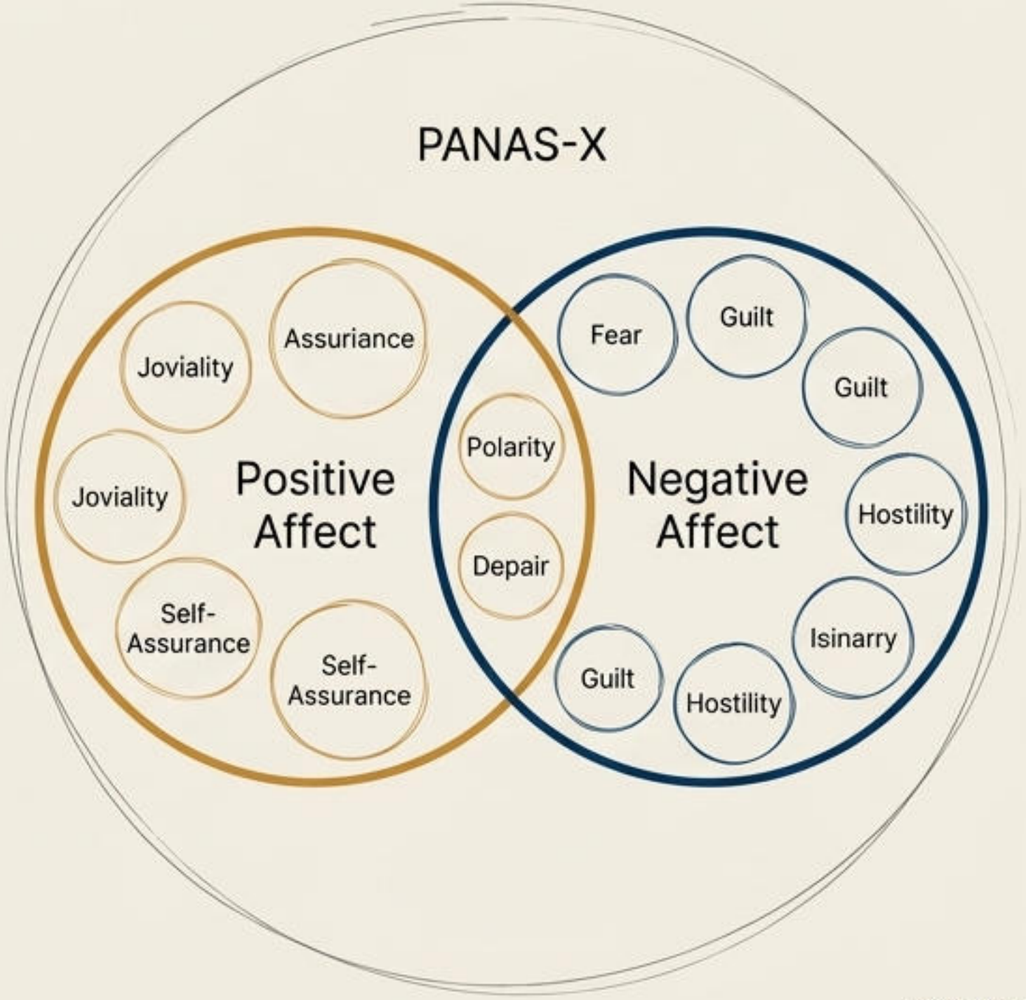
The PANAS-X: Expanding Measurement

The original 20-item PANAS was later expanded to the 60-item PANAS-X. This expanded form retains the higher-order Positive and Negative Affect scales but also provides scores for 11 specific affective states, allowing for measurement at two levels of the affective hierarchy.

The 20 terms comprising the two main scales were carefully selected to be relatively pure markers of either PA or NA.

- **Positive Affect Descriptors:** include terms like *active*, *interested*, *excited*, *strong*, *proud*, *enthusiastic*, *inspired*, *determined*, *attentive*, and *alert*. These words of *attentive*, and *alert*. These words capture a state of approach and engagement.
- **Negative Affect Descriptors:** include terms like *distressed*, *upset*, *guilty*, *scared*, *hostile*, *irritable*, *ashamed*, *nervous*, *jittery*, and *afraid*. These words capture a state of aversion and distress.

This instrument provides a robust and comprehensive tool for quantifying the complex landscape of human mood.



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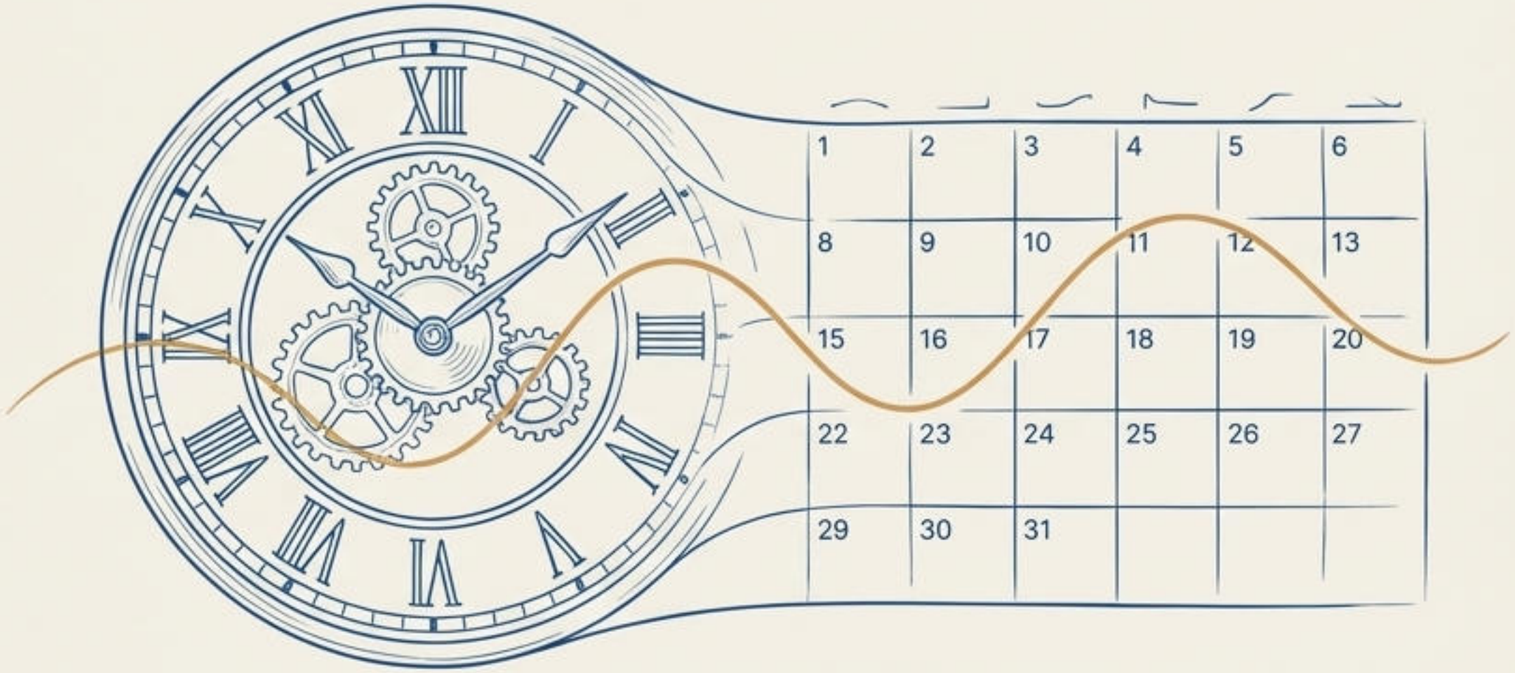
Biological and Temporal Rhythms

Mood is not a static condition; it is in constant, subtle flux. These fluctuations are not random but are often tied to systematic, cyclical changes in our environment and our internal biology. Research has consistently shown that our affective states follow predictable temporal patterns over the course of a day and across the week.

Understanding these rhythms is crucial for comprehending the baseline dynamics of mood, independent of specific life events. These patterns are governed by internal pacemakers, known as circadian rhythms, as well as by the social and structural schedules of our lives. We will now explore these diurnal and weekly cycles.



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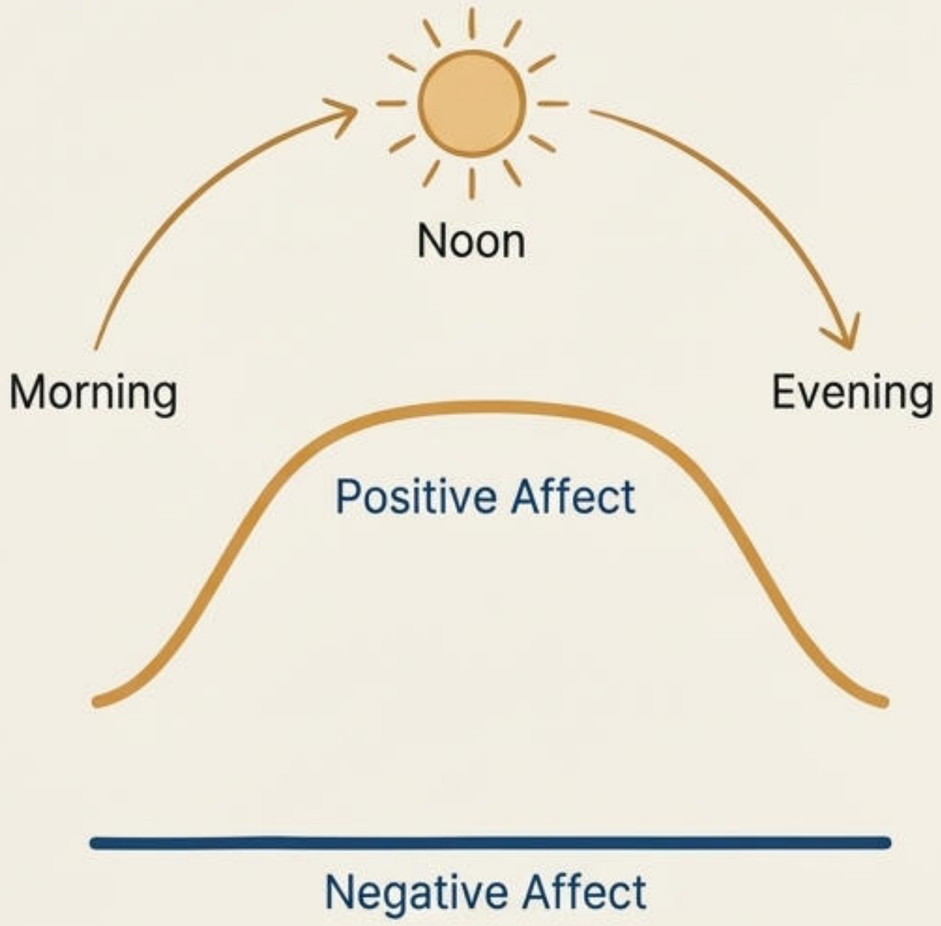
Diurnal Rhythms: Mood Across the Day

Studies measuring affect at multiple points throughout the day reveal a consistent diurnal pattern, particularly for positive affect.

Clark et al. (1989) found that **Positive Affect (PA)** follows a distinct curve:

- It starts low in the early morning.
- It rises rapidly from around 9:00 a.m. to a peak around noon.
- It remains relatively stable and high from noon until approximately 9:00 p.m.
- It then declines sharply in the late evening.

In contrast, **Negative Affect (NA)** exhibits remarkable stability. Across most studies, ratings of negative mood remain low and relatively flat throughout the day, showing no systematic diurnal variation. This suggests that while positive feelings are tightly coupled with our daily cycle of energy and activity, negative feelings are more likely to be elicited by specific, aperiodic stressors.



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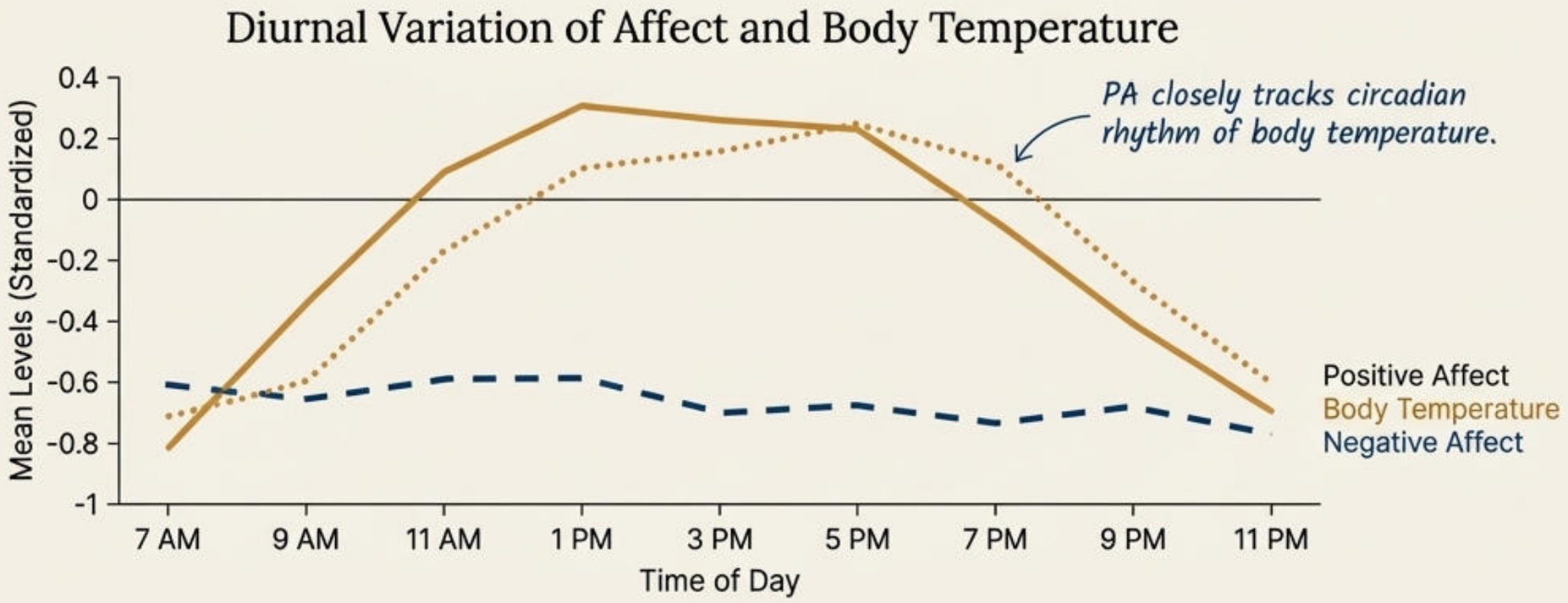
Correlation with Circadian Rhythms

The daily rhythm of positive affect is not merely a social phenomenon; it is closely linked to our underlying biological processes. As demonstrated by Watson et al. (1999), the diurnal curve of positive mood closely parallels the **circadian rhythm** of core body temperature.

Both positive affect and body temperature are low in the early morning, rise throughout the day, and fall in the late evening. This biological rhythm refers to the 24-hour cycle of physiological processes, including the **sleep/wake cycle**. The strong correlation suggests that our experience of positive, energetic feelings is deeply rooted in our fundamental **biological clocks**, reflecting daily fluctuations in physiological arousal and metabolic activity.



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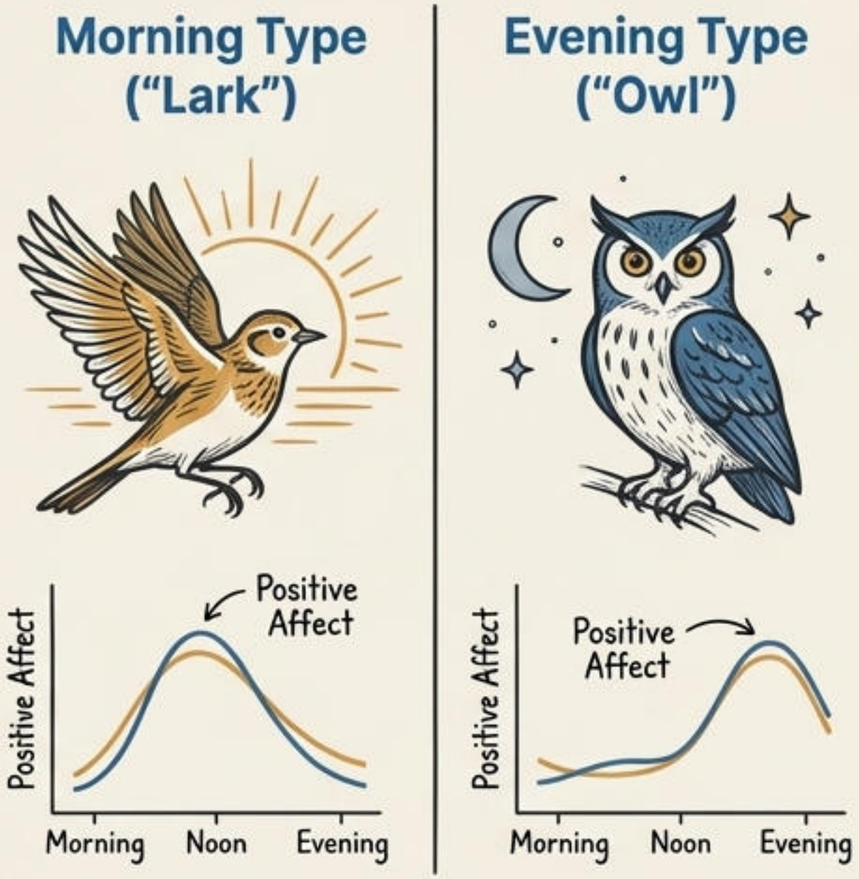
Individual Differences: Morningness–Eveningness

The general diurnal pattern of mood is not universal but is moderated by a stable personality trait known as **morningness–eveningness**. This dimension reflects the time of day an individual functions best and feels most alert.

Morning Types ("Larks"): Prefer and perform best earlier in the day. Their positive affect rises earlier and peaks in the late morning. They are often more cheerful and active in the morning compared to the evening.

Evening Types ("Owls"): Prefer and perform best later in the day. Their rise in positive affect is delayed, and they may not reach their peak mood until early evening.

The *Morningness–Eveningness Questionnaire* is a self-report scale used to measure an individual's position on this chronotype dimension. This trait highlights that our mood rhythms are an interaction between a general biological template and individual trait-based variations.



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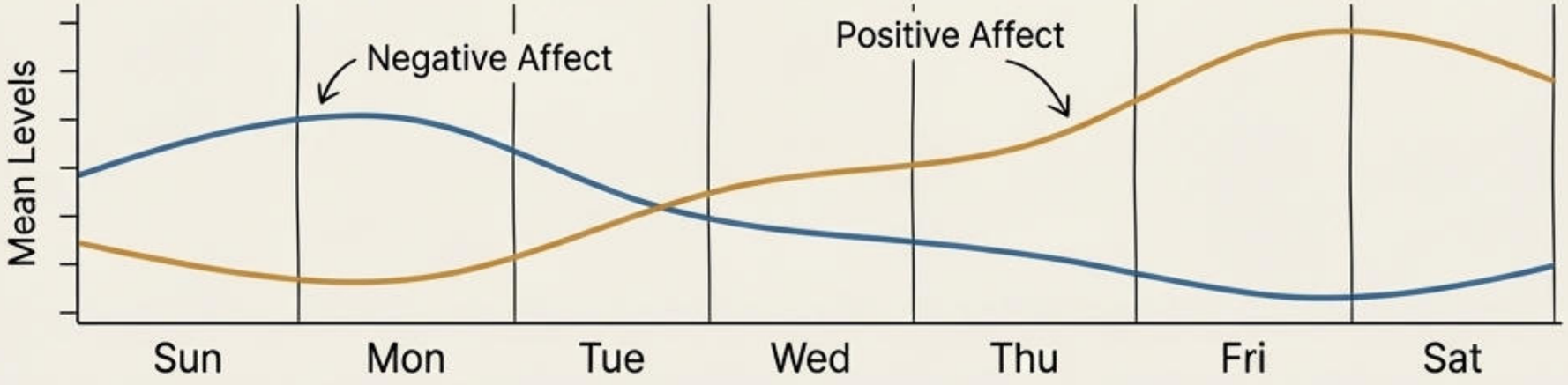
Weekly Mood Cycles

Just as mood fluctuates across the day, it also follows a seven-day cycle, largely dictated by the social convention of the workweek and weekend. Studies tracking daily mood ratings across many weeks reveal a clear and replicable pattern:

- **Positive Affect** is lowest on Sunday and Monday. It then begins a steady ascent through the week, peaking on Friday and Saturday.
- **Negative Affect** shows a reverse trend. It often rises from Sunday to Monday, then steadily declines for the remainder of the week, reaching its lowest point on Friday and Saturday.

The period from Thursday through Saturday, where positive affect is high and negative affect is low, has been termed the “subjective weekend” for college students (Watson, 2000), representing a peak in overall well-being.



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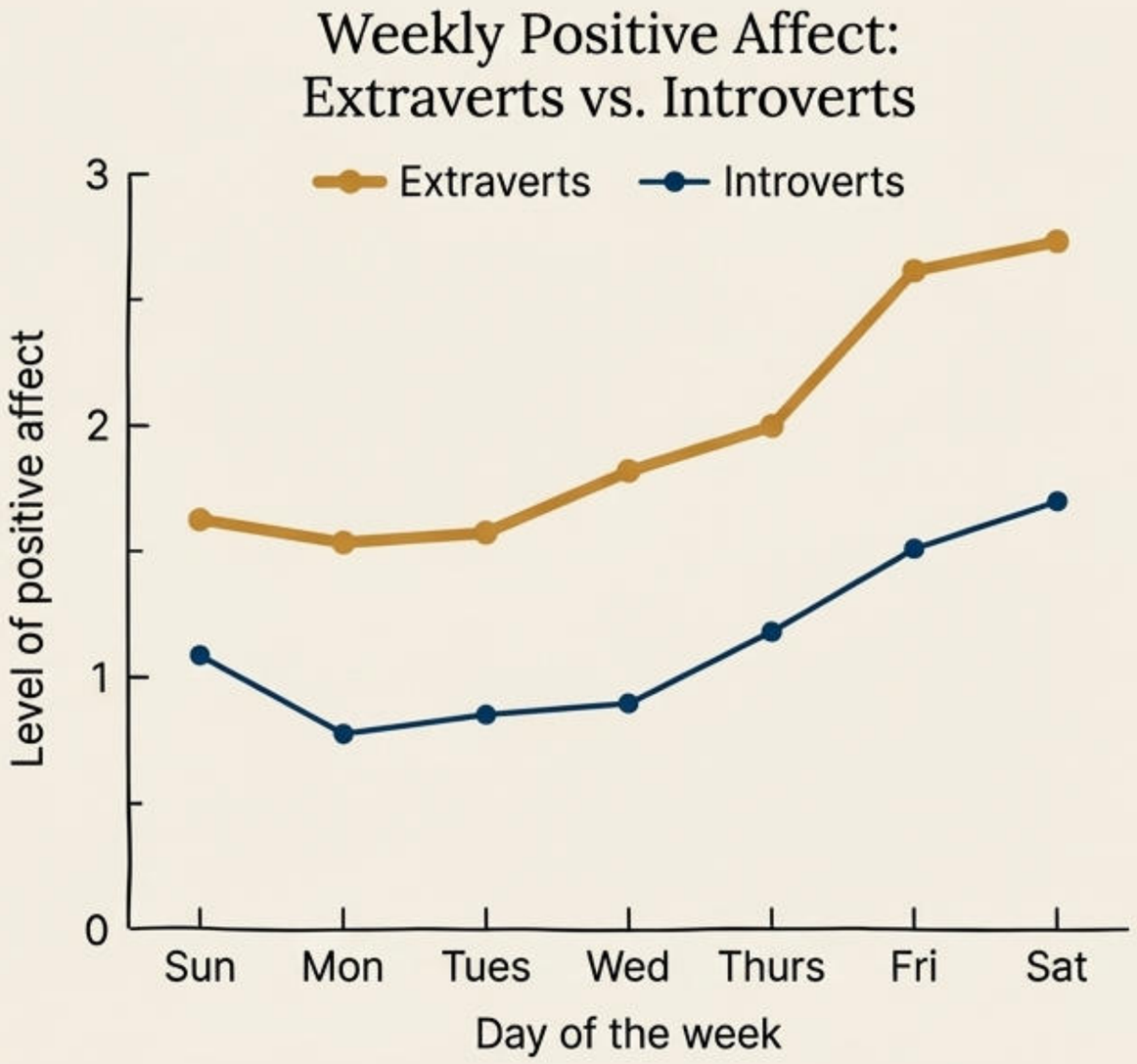
Personality as a Moderator: Extraversion and Introversion

The weekly mood cycle is also moderated by personality. Larsen and Kasimatis (1990) demonstrated that the trait of **extraversion** significantly influences the level and trajectory of positive affect across the week.

As shown in the data, extraverts consistently report higher levels of positive affect than introverts on every day of the week. Furthermore, the rise in positive affect as the weekend approaches is significantly steeper for extraverts. This finding aligns with theories linking extraversion to greater sensitivity to rewards and positive emotional stimuli, which are often more abundant during the social activities of the weekend. Introverts experience a similar weekly pattern, but it is more muted in both its baseline level and its weekend ascent.



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Environmental and Lifestyle Determinants

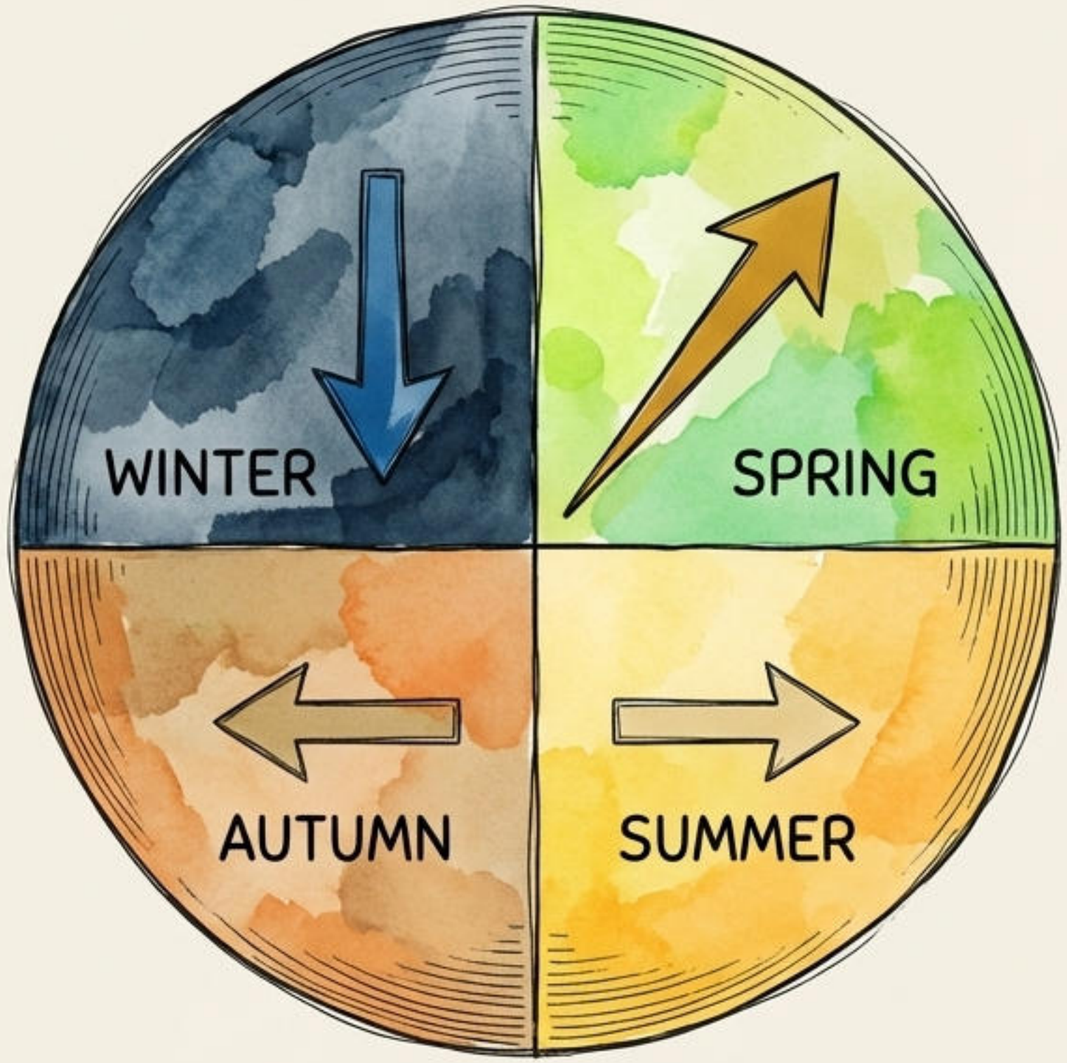
Seasonal Variations and Mood

Beyond daily and weekly cycles, moods can be influenced by longer-term environmental shifts, most notably the changing of the seasons. This effect is particularly pronounced in regions with significant variation in daylight and weather.

The Seasonal Pattern Assessment Questionnaire (SPAQ) is used to measure individuals' self-reported changes in mood and behavior with the seasons.

- **Winter**: Consistently emerges as the 'least favorite' season, associated with lower mood, energy, and social interest for a significant portion of the population.
- **Spring**: Is typically associated with a 'revival of happiness.' As the weather warms, vegetation returns, and daylight hours lengthen, there is a corresponding upswing in positive mood.

These slow, pervasive environmental changes can create a long-term backdrop that subtly shapes our baseline affective state.



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Daily Weather and the Law of Hedonic Contrast

While the broad changes of seasons have a clear impact, demonstrating mood changes from daily weather variations (temperature, sunlight, wind) has proven more complex. The effects are often small but can be sime significant under certain conditions.

- **Moderating Factors:** The influence of weather is stronger for individuals who spend more time outdoors. Keller et al. (2005) found that for those outside more than 30 minutes, mood improved with higher temperatures and more sunlight.

- **The Law of Hedonic Contrast:** This principle suggests that the enjoyment of a pleasant stimulus is amplified when it is preceded by an unpleasant one. A warm, sunny day is perceived as more mood-enhancing after a long stretch of cold, rainy days than it is during a consistent heatwave. Spring's positive effect, therefore, is partly due to its contrast with the preceding winter.

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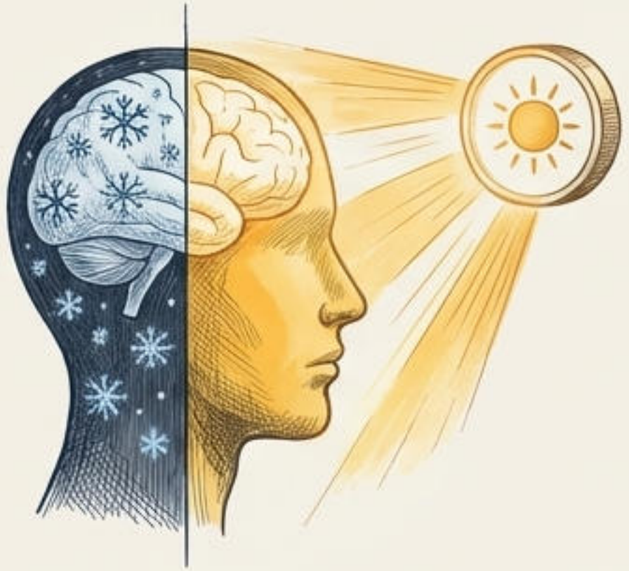
Clinical Considerations: Seasonal Affective Disorder (SAD)

For some individuals, seasonal mood changes are not mild but constitute a clinical condition known as **Seasonal Affective Disorder (SAD)**. SAD is a form of depression with a recurring seasonal pattern.

Characteristics:

- Typically begins in autumn or winter and remits in spring.
- Symptoms include sadness, anhedonia, decreased activity, anxiety, and irritability.
- Atypical depressive symptoms are common, such as hypersomnia (daytime tiredness, oversleeping) and increased appetite/weight gain.

The prevailing theory links SAD to decreased sunlight exposure in winter, which may disrupt circadian rhythms and neurotransmitter systems.



Treatments:

- **Light Therapy**: Daily exposure to a bright, full-spectrum light box is a first-line treatment.
- **Geographical Relocation**: Moving to latitudes closer to the equator with more consistent daylight can alleviate symptoms.

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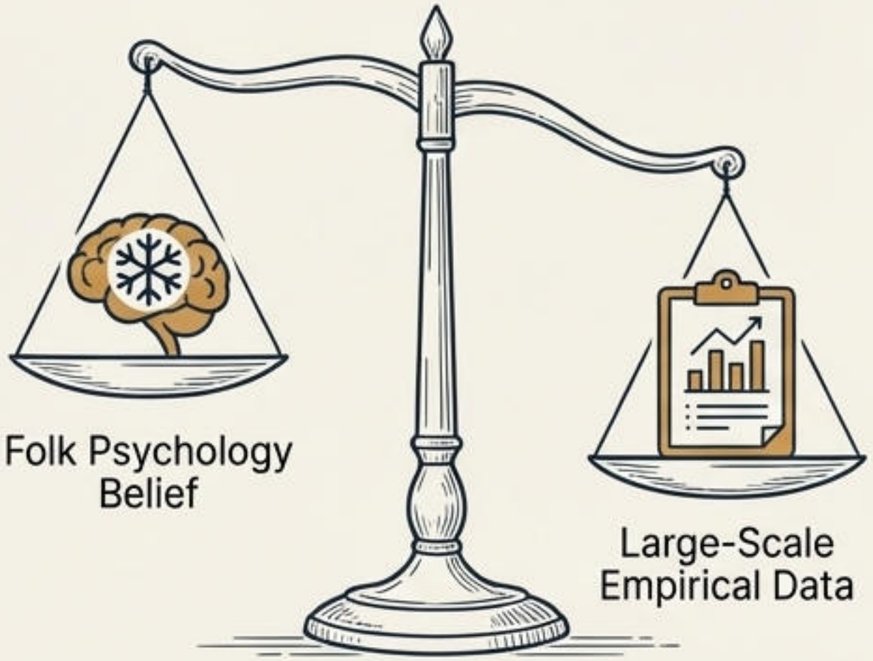
Critiques and the Role of Folk Psychology

While the concept of SAD is widely accepted, its prevalence and causal link to severe depression have been challenged by recent large-scale empirical studies.

Traffanstedt et al. (2016) analyzing data from over 34,000 U.S. adults, found no association between diagnoses of major depression and the season, latitude, or amount of sunlight exposure. This suggests that while milder "winter blues" and seasonal variations in mood are real, major depression as a clinical disorder may not have a seasonal basis.

The authors conclude that the strong belief in winter depression may be more rooted in **folk psychology**—a culturally shared, intuitive belief—than in robust empirical evidence. The key distinction is critical: "being depressed during winter is not evidence that one is depressed *because of* winter." This highlights the importance of differentiating subjective seasonal mood shifts from clinically significant depressive disorders.



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The Impact of Sleep Deprivation on Affect

Beyond external factors, fundamental lifestyle behaviors like sleep have a profound and immediate impact on mood. While the experience of being "cranky" after a poor night's sleep is common, experimental research reveals a specific and asymmetric effect on the dimensions of affect.

In a key study by Talbot et al. (2010), researchers investigated the effects of acute sleep deprivation on the moods of participants.

- **Procedure:** Participants first had a baseline night of adequate sleep, followed by a night of severe sleep deprivation (e.g., 2 hours of sleep). Mood was measured using the PANAS after both conditions, followed by recovery sleep.
- **Hypothesis:** Sleep loss would disrupt affective regulation, but not necessarily in a uniform way. It might degrade positive feelings more than it amplifies negative ones.

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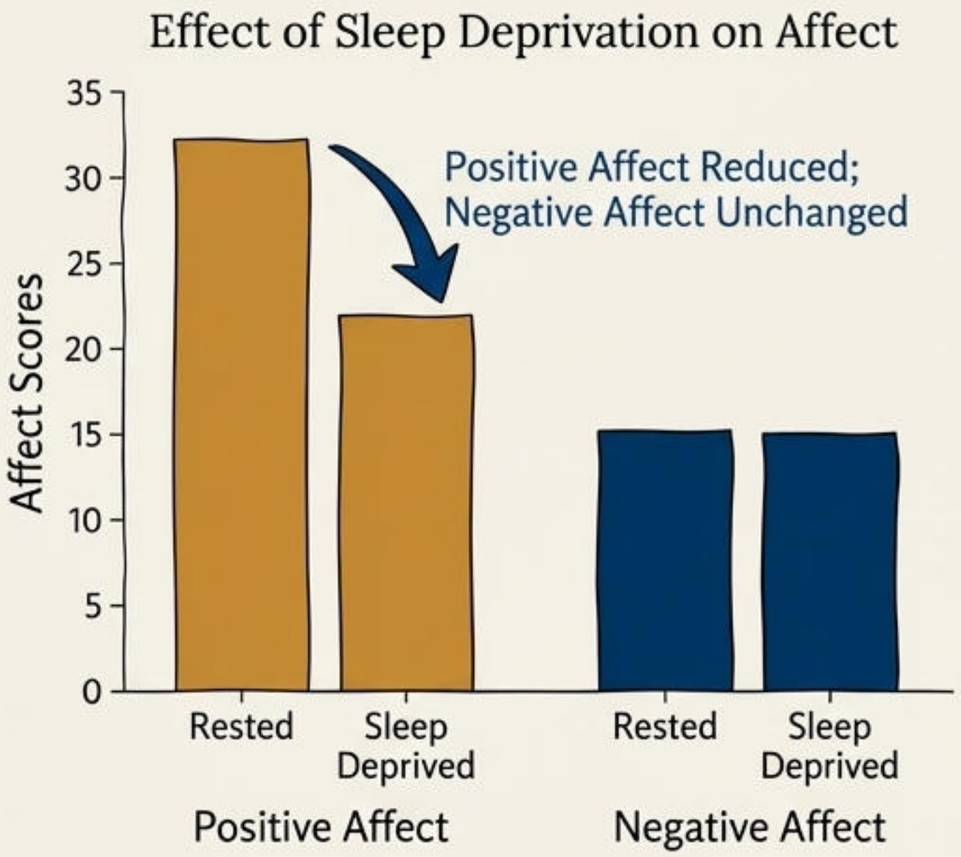
Differential Effects on Positive and Negative Affect

The findings from Talbot et al. (2010) were striking and have been replicated. Sleep deprivation does not simply make people feel "bad" in a general sense.

- It caused a significant **reduction in positive affect** (#003366). Participants felt less active, enthusiastic, and energetic when sleep-deprived compared to when they were rested.
- It had **no significant effect on negative affect** (#003366). Levels of irritability, distress, and nervousness remained relatively stable and did not increase with sleep loss.

This demonstrates that a primary consequence of sleep deprivation is the blunting of positive emotional experience, **rather than an amplification of negative feelings.** Following two nights of recovery sleep, positive affect scores returned to their baseline levels, confirming the causal role of sleep in maintaining our capacity for positive mood.



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